EXECUTIVE BUSINESS REPORT

End-to-End Sales Forecasting & Demand Intelligence System

Executive Summary

This project presents a comprehensive Sales Forecasting and Demand Intelligence System developed using four years of retail sales data from the Superstore Sales dataset. The objective was to analyze historical sales patterns, forecast future product demand, detect unusual sales behavior, segment products based on demand characteristics, and present the findings through an interactive Streamlit dashboard. Three forecasting approaches—SARIMA, Facebook Prophet, and XGBoost—were implemented and compared using multiple evaluation metrics. The project demonstrates how data analytics and machine learning can support inventory planning, procurement decisions, and overall business strategy by providing accurate forecasts and actionable insights.

Key Findings from Exploratory Data Analysis and Forecasting

The exploratory data analysis revealed several important business insights regarding sales performance and customer demand.

- Technology products generated the highest overall sales revenue among all product categories, making them the primary contributor to business growth.
- Monthly sales exhibited clear seasonal patterns, with noticeable increases during festive seasons and year-end shopping periods.
- Regional sales analysis showed that certain regions maintained more consistent sales performance than others, indicating differences in customer demand and market stability.
- Shipping time analysis showed that the average delivery duration remained relatively consistent across regions, suggesting an efficient logistics process.
- Time series decomposition confirmed the presence of both long-term sales trends and recurring seasonal patterns, making the data suitable for forecasting models.

Three forecasting models were developed and evaluated:

- SARIMA successfully captured seasonal behavior and generated reliable forecasts with statistical confidence intervals.
- Facebook Prophet effectively modeled long-term trends and yearly seasonality while requiring minimal parameter tuning.

- XGBoost used lag variables, rolling averages, and calendar-based features to learn complex nonlinear demand patterns.

Based on the evaluation metrics (MAE, RMSE, and MAPE), the **Facebook Prophet model demonstrated the best overall forecasting performance** and is recommended as the production forecasting model for future sales prediction.

Three-Month Sales Forecast

The forecasting models indicate that overall sales are expected to remain stable over the next three months while continuing to follow historical seasonal demand patterns.

The Prophet model predicts moderate sales growth with confidence intervals that provide a realistic range of possible future sales values. These confidence intervals help decision-makers estimate inventory requirements while accounting for uncertainty in customer demand.

The forecast suggests that the organization should maintain sufficient inventory levels during upcoming high-demand periods to minimize stock shortages while avoiding unnecessary overstocking.

Top Three Anomalies Detected and Their Likely Causes

The anomaly detection phase used both Isolation Forest and Z-Score methods to identify unusual sales behavior.

Anomaly 1: A significant sales spike was detected during one of the year-end months. This increase is likely associated with festive shopping seasons, promotional campaigns, or holiday discounts that temporarily increased customer demand.

Anomaly 2: An unusually low sales period was observed during another month. This decline may have resulted from supply chain disruptions, reduced customer demand, inventory shortages, or external market conditions.

Anomaly 3: A second high-sales anomaly was identified where demand exceeded normal seasonal expectations. This unusual increase could indicate successful marketing campaigns, bulk customer purchases, or special promotional events.

The comparison between Isolation Forest and Z-Score demonstrated that while both methods identified several common anomalies, Isolation Forest was more effective at detecting complex abnormal patterns because it considers multidimensional relationships rather than relying solely on statistical deviations.

Product Demand Segmentation and Stocking Strategy

K-Means Clustering was applied to group products according to their demand characteristics using features such as sales volume, sales variability, growth rate, and average order value.

Principal Component Analysis (PCA) was used to visualize the resulting clusters.

The clustering process identified three major product demand segments:

High Volume, Stable Demand

These products consistently generate strong sales and should receive the highest inventory priority. Frequent replenishment and safety stock should be maintained to prevent stock shortages.

Moderate Demand Products

These products exhibit stable but moderate demand. Balanced inventory management with regular demand monitoring is recommended to maintain operational efficiency while controlling inventory costs.

Low Volume or High Volatility Products

These products experience relatively lower or unpredictable demand. Inventory should be maintained cautiously with smaller reorder quantities and frequent performance monitoring to reduce excess stock and storage expenses.

This segmentation enables businesses to allocate inventory more efficiently while improving warehouse utilization and reducing inventory holding costs.

Business Recommendations

1. Adopt Facebook Prophet for Production Forecasting

Based on the comparative evaluation of SARIMA, Prophet, and XGBoost, Prophet produced the most accurate forecasts and should be implemented as the primary forecasting model for monthly demand planning.

2. Strengthen Inventory Planning During Seasonal Peaks

Historical sales analysis identified recurring seasonal demand patterns. Increasing inventory levels before high-demand periods can reduce stock shortages, improve customer satisfaction, and maximize sales opportunities.

3. Continuously Monitor Sales Anomalies

The anomaly detection system should be integrated into routine business monitoring to quickly identify unexpected demand changes, promotional impacts, operational issues, or data quality problems. Early detection enables faster business response and better decision-making.

Risk and Limitation

Although the forecasting system achieved strong predictive performance, the models rely primarily on historical sales data. External variables such as holidays, competitor pricing, economic conditions, weather, inflation, customer sentiment, and marketing campaigns were not incorporated into the forecasting models. Including these additional business factors in future versions could further improve forecasting accuracy and decision-making.

Conclusion

The End-to-End Sales Forecasting and Demand Intelligence System successfully combines exploratory data analysis, time series forecasting, machine learning, anomaly detection, product demand segmentation, and interactive visualization into a unified business intelligence solution. The project demonstrates how advanced analytics can support strategic inventory planning, procurement optimization, and sales forecasting in retail organizations. By leveraging accurate forecasting models and data-driven insights, businesses can reduce operational costs, improve inventory efficiency, and make more informed decisions. The deployed Streamlit dashboard further enhances accessibility by allowing business users to interactively explore sales trends, forecasts, anomalies, and product demand segments without requiring technical expertise.